

ACTION RECOVERY FROM PROJECTIVE MATRIX CURVES

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ABSTRACT. We study recovery of an unordered action spectrum from normalized matrix observations with two unknown stochastic channels. For every fixed latent rank, three clocks identify every nonconstant spectrum with full-rank channels, whereas two clocks admit a nontrivial exact fibre. A one-sided inverse estimate holds against the full closed model, including rank-deficient competitors, without a lower bound on individual action gaps. Its conditioning separates the product of the two smallest channel singular values from the observable contrast of the projective curve. Thus partial action collisions and collapse of the entire spectrum have different consequences. Residual confidence sets are honest without supplied rank or contrast floors. Explicit positive families with unchanged marginals, and a second family with a collapsing spectrum, give matching risk and confidence-diameter orders in every latent rank. The lower bounds also hold under adaptive clock selection. The binary determinant law is recovered as a separated-spectrum case.

1. INTRODUCTION

A selective observation can erase the total mass of an experiment without erasing the relative frequencies of two conditionally independent readings. The resulting datum is a probability matrix. When an unobserved component with action ξ_b contributes a factor $T - \xi_b$ at clock T , the matrix is a rational curve with an unknown common pole. The action spectrum is a feature of this curve, not of a particular naming of the hidden components.

The exact matrix-pencil inverse is well understood when the channels are well conditioned and the actions are separated. The question here is different: how much information about the *actions* remains when the channels approach rank loss, some actions coincide, or the whole clock curve approaches a constant? These limits should not be combined into an unspecified inverse-function constant. Nor should conditioning for recovering eigenvectors be charged to a problem whose target is only the unordered spectrum.

We prove one estimate which separates the two relevant losses. The first is bilinear: a weak direction in each readout channel produces the product of the two weaknesses. The second is projective: a nearly constant normalized curve conceals the location of its common pole. A partial collision of actions causes neither loss by itself. The same estimate yields exact identification, honest confidence regions, and minimax upper bounds. Two explicit families show that its two singular orders cannot in general be improved.

1.1. The experiment. Fix integers $2 \leq k \leq \min(m_1, m_2)$, real numbers $L < D < T_1 < T_2 < T_3$, and $0 < \alpha_* < 1/k$. Let $\mathcal{K}$ be the compact set of parameters

$$\omega = (U, V, \alpha, \xi), \quad U \in \mathbb{R}^{m_1 \times k}, \quad V \in \mathbb{R}^{m_2 \times k},$$

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where the columns of U, V are probability vectors, $\alpha_b \geq \alpha_*$, $\sum_b \alpha_b = 1$, and $L \leq \xi_1 \leq \dots \leq \xi_k \leq D$. Rank-deficient channels and repeated actions are included. Define

$$\begin{aligned} h &= \sum_b \alpha_b \xi_b, & p_b(T) &= \frac{\alpha_b(T - \xi_b)}{T - h}, \\ P_\omega(T) &= U \operatorname{diag}(p_b(T)) V^\top, & \mathcal{P}(\omega) &= (P_\omega(T_1), P_\omega(T_2), P_\omega(T_3)). \end{aligned} \quad (1.1)$$

Every $P_\omega(T)$ is a probability matrix. All weights $p_b(T)$ have a positive lower bound on $[T_1, T_3]$. The normalization of raw matrices $c(T)U \operatorname{diag}(a_b(T - \xi_b))V^\top$ gives (1.1), with $\alpha_b = a_b / \sum_c a_c$, for every unknown positive scalar detector $c(T)$. Moreover

$$\log \frac{p_b(T)}{p_c(T)} = \log \frac{\alpha_b}{\alpha_c} + \log(T - \xi_b) - \log(T - \xi_c).$$

Thus the original logarithmic action law is retained. Componentwise clock-dependent detectors are a different experiment.

The target is the spectrum with multiplicity, metrized by

$$d_\infty(\xi, \xi') = \min_{\pi \in \mathfrak{S}_k} \max_b |\xi_b - \xi'_{\pi(b)}| = \max_b |\xi_b - \xi'_b|.$$

The last equality holds for the displayed increasing order. We use $\|\mathcal{P} - \mathcal{P}'\|_* = \max_j \|P_j - P'_j\|_F$. All unspecified constants depend only on the fixed dimensions, intervals, weight floor and clocks.

Put

$$\begin{aligned} A &= U \operatorname{diag}(\alpha_b) V^\top, & B &= U \operatorname{diag}(\alpha_b(h - \xi_b)) V^\top, \\ \kappa(\omega) &= \sigma_k(U) \sigma_k(V), & \gamma(\omega) &= \|B\|_F, \quad \vartheta(\omega) = \frac{\kappa \gamma}{\kappa + \gamma}. \end{aligned} \quad (1.2)$$

Set $\vartheta = 0$ when either factor vanishes. In particular ϑ is continuous on the closed model. The identity

$$P_\omega(T) = A + \frac{B}{T - h} \quad (1.3)$$

is the starting point of the argument.

Theorem 1.1 (Projective spectrum and singular modulus). *If $\vartheta(\omega) > 0$, then for every $\omega' \in \mathcal{K}$,*

$$d_\infty(\xi, \xi') + |h - h'| \leq C \min \left\{ 1, \frac{\|\mathcal{P}(\omega) - \mathcal{P}(\omega')\|_*}{\vartheta(\omega)} \right\}. \quad (1.4)$$

There is no rank, contrast, or individual action-gap assumption on the competitor. No individual action-gap constant enters C . Consequently three clocks identify the spectrum and h whenever the channels have full column rank and not all actions coincide. Two clocks do not identify them, even with fixed positive full-rank channels and interior, distinct actions.

If $\xi_k - \xi_1 \geq \Delta > 0$, then $\vartheta(\omega) \geq c_\Delta \kappa(\omega)$, so (1.4) holds with κ in place of ϑ and a constant depending additionally on Δ , but on no smaller gap.

The theorem extends the earlier arbitrary-rank exact clock-pencil inverse by a uniform, one-sided singular estimate. At simple spectra it also identifies the channels and relative weights up to a common permutation; at repeated spectra these factors can have additional gauges. The estimate concerns the spectrum, and does not assert uniform recovery of those factors through an eigenvalue collision.

For an intrinsic formulation, define on the model image

$$\vartheta_*(P) = \max\{\vartheta(\omega) : \mathcal{P}(\omega) = P\}. \quad (1.5)$$

Compactness makes the maximum attainable. This is invariant under all latent representations of the same datum, not merely under permutation. If it is positive, Theorem 1.1 shows that all

representatives have the same spectrum and pole. The estimate then holds with $\vartheta_*(P)$, by using a maximizing representative. For a simple spectrum the stochastic spectral factors determine κ up to permutation, and hence $\vartheta_* = \vartheta$. For repeated spectra (1.5) is a compact factorization problem; no unproved closed formula is assumed.

1.2. Statistical consequence. At each of the three clocks observe n independent draws from the $m_1 m_2$ -cell law, and let $N = 3n$. Section 4 constructs a residual estimator and an honest confidence region over the entire closed model, without supplying either conditioning factor. For each true observation P with $\vartheta_*(P) > 0$, their squared risk and expected diameter satisfy

$$\mathbb{E} d_\infty(\hat{\xi}, \xi)^2 \leq C \min\{1, (N\vartheta_*(P)^2)^{-1}\}, \quad \mathbb{E} \text{diam}_\infty \mathcal{C}_\xi \leq C_\zeta \min\{1, (\sqrt{N}\vartheta_*(P))^{-1}\}. \quad (1.6)$$

These orders are minimax sharp on the classes $\{\omega : \vartheta_*(\mathcal{P}(\omega)) \geq s\}$ for all sufficiently small $s > 0$. They are attained on two different positive families: one with weak channel directions and fixed action spread, and one with full-rank channels and a collapsing action spread. The same lower bounds hold for adaptive selection of clocks in $[T_1, T_3]$.

1.3. Predecessors and the contribution. Observable diagonalization and latent-structure identification are classical; see Allman–Matias–Rhodes [1] and Bonhomme–Jochmans–Robin [3]. In this experiment a clock is a controlled covariate, not an additional conditionally independent readout. We use the usual rank-one spectral decomposition, rather than claim a new general tensor-identifiability theorem. The exact three-clock reconstruction was already available in the earlier clock-pencil derivation. The present assertion is its gap-free singular modulus and its finite-experiment lower bounds.

Identification modulo nuisance transformations has a general bilinear formulation in Li–Lee–Bresler [8]. Here the normalization nuisance has a specific one-parameter two-clock fibre, and three observations remove it. The perturbation proof below is related to the classical matrix-pencil perspective of Bauer–Fike [2]; we include the required eigenvalue argument. Applying a condition number to an observed inverse and then another to its eigenvectors would not establish the stated product law. The essential step is to keep the pole perturbation diagonal after multiplication by the *true* two channel inverses.

The relation between moduli and inference is classical [5, 4]; weak identification and nonshrinking honest sets have important precedents in [6]. Ho–Nguyen [7] study singularity structures and rates for finite mixtures. The calculation here is for a controlled categorical experiment with unknown channels. Partial coalescence of action values does not cause a slower action-spectrum rate: the full matrix curve still retains the spectrum. Loss of a channel direction and loss of the whole projective contrast do cause singular rates, as the two exact families demonstrate. This distinction is a statement about the observation map and the chosen target, not about finite mixtures in general.

2. THE PROJECTIVE FIBRE AND ITS THIRD-CLOCK RESOLUTION

Lemma 2.1 (Pole recovery). *If $\gamma > 0$, set $X = P_1 - P_2$, $Y = P_2 - P_3$. Then*

$$\frac{\langle X, Y \rangle_F}{\|Y\|_F^2} = \frac{T_2 - T_1}{T_3 - T_2} \frac{T_3 - h}{T_1 - h}, \quad c\gamma \leq \|Y\|_F \leq C\gamma. \quad (2.1)$$

If $\delta = \|\mathcal{P} - \mathcal{P}'\|_ \leq c_0\gamma$ for a sufficiently small fixed c_0 , the candidate also has $\gamma' > 0$, and*

$$|h - h'| \leq C\delta/\gamma. \quad (2.2)$$

Proof. Subtract (1.3) at consecutive clocks. The differences are scalar multiples of B , which gives (2.1). All scalar multipliers and their relevant reciprocals are bounded because $h \in [L, D]$. If $\delta \leq c_0\gamma$, then $\|Y'\|_F \geq \|Y\|_F - 2\delta \geq c\gamma/2$; in particular B' is nonzero. The

difference of the two ratios is at most $C\delta/\gamma$: expand their numerators and denominators, using $\|X\|_F, \|Y\|_F, \|X'\|_F, \|Y'\|_F \leq C\gamma$ and the lower bounds on the denominators. The derivative of the right side of (2.1) is

$$\frac{T_2 - T_1}{T_3 - T_2} \frac{T_3 - T_1}{(T_1 - h)^2},$$

which is positive and bounded below on $[L, D]$. This proves (2.2). $\square$

Once h is recovered, two values of $(T - h)P(T)$ give the affine polynomial

$$K(T) = U \operatorname{diag}(\alpha_b(T - \xi_b)) V^\top. \quad (2.3)$$

Its reduced generalized eigenvalues are the actions, with multiplicity. Section 3 proves this and its stability without first recovering individual eigenvectors.

Proposition 2.2 (The exact two-clock fibre). *Fix an interior parameter with positive full-rank channels and distinct actions. Its two-clock fibre contains a smooth nonconstant curve with these same channels. With $H = (T_1 - h)(T_2 - h)$ and sufficiently small real u , this curve is*

$$\begin{aligned} h_u &= h + u, \\ \alpha_{b,u} &= \alpha_b \left(1 + \frac{u(h - \xi_b)}{H} \right), \\ \xi_{b,u} &= \frac{\xi_b + u\{1 + (h - \xi_b)(T_1 + T_2 - h)/H\}}{1 + u(h - \xi_b)/H}. \end{aligned} \quad (2.4)$$

Both observed matrices are exactly unchanged. Moreover

$$\left. \frac{d\xi_{b,u}}{du} \right|_{u=0} = \frac{(T_1 - \xi_b)(T_2 - \xi_b)}{H} > 0. \quad (2.5)$$

Proof. Let $q_b(T)$ be the affine interpolant of $p_b(T_1), p_b(T_2)$. Direct calculation gives

$$q_b(T) = \alpha_b \left[1 + (h - \xi_b) \frac{T_1 + T_2 - h - T}{H} \right]. \quad (2.6)$$

The affine function $\alpha_b(T - \xi_b) - uq_b(T)$ has slope $\alpha_{b,u}$ and root $\xi_{b,u}$. At T_j , $j = 1, 2$, it equals $(T_j - h - u)p_b(T_j)$. Summing over b gives the polynomial $T - h - u$. Thus $\sum_b \alpha_{b,u} = 1$ and $\sum_b \alpha_{b,u} \xi_{b,u} = h + u$, and the two normalized weights are unchanged. Positivity, the weight floor, distinctness and action bounds persist for small u by the interior assumptions. Differentiation gives (2.5). $\square$

The obstruction is present even when the channels are fixed. It is therefore a normalization fibre, not a failure of a spectral algorithm.

3. A GAP-FREE ONE-SIDED INVERSE

We first record an elementary spectral fact. The use of a multiset rather than individually labelled eigenvectors is essential.

Lemma 3.1 (Matching a perturbed real diagonal spectrum). *Let $D_0 = \operatorname{diag}(t_1, \dots, t_k)$ with increasing real t_b . If a real matrix $D_0 + E_0$ has real eigenvalues $t'_1 \leq \dots \leq t'_k$, counted with algebraic multiplicity, then*

$$\max_b |t_b - t'_b| \leq (2k - 1) \|E_0\|_{\text{op}}. \quad (3.1)$$

Repeated eigenvalues in either spectrum are allowed.

Proof. Write $\varepsilon = \|E_0\|_{\text{op}}$. For every $0 \leq s \leq 1$, the spectrum of $D_0 + sE_0$ is in the union of the complex disks of radius ε about t_b : outside this union the diagonal resolvent has norm less than ε^{-1} , so a Neumann series proves invertibility. Use slightly larger disks first, choosing their radius so distinct connected components have disjoint boundaries. The number of eigenvalues in each component is constant along the homotopy, by the argument principle for the characteristic polynomial. It is the number of its centres, counted with multiplicity. A component with r centres has diameter between extreme centres at most $2(r-1)\varepsilon$, in the limiting radius. A root in the component is consequently at distance at most $(2r-1)\varepsilon$ from every one of those centres. Match within each component and let the additional radius decrease to zero. For real ordered spectra an increasing matching minimizes the maximum distance: an inverted pair can be uncrossed without increasing that distance. This proves (3.1). $\square$

Proof of Theorem 1.1. It suffices first to consider $\delta \leq c_0\vartheta$. Let $L_U = U^\dagger$ and $R_V = (V^\dagger)^\top$ be the true channel inverses. They are used only in the proof, not supplied to the estimator. Their norms have product κ^{-1} . Let $K'(T) = (T - h')P_{\omega'}(T)$, an affine matrix polynomial even when the candidate channels are singular, and write

$$M(T) = L_U K'(T) R_V.$$

At T_j , $j = 1, 2$,

$$\begin{aligned} M(T_j) &= \text{diag}(\alpha_b(T_j - \xi_b)) - (h' - h) \text{diag}(p_b(T_j)) + F_j, \\ F_j &= (T_j - h')L_U(P'_j - P_j)R_V, \quad \|F_j\|_{\text{op}} \leq C\delta/\kappa. \end{aligned} \quad (3.2)$$

Affinely interpolate this identity. The coefficient norms of the interpolant of F_1, F_2 are bounded by $C\delta/\kappa$. The pole term is diagonal, with scalar affine entries q_b in (2.6); their coefficients are uniformly bounded. Hence

$$M(T) = (D_\alpha + E_1)T - D_\alpha D_\xi + E_0, \quad \|E_0\|_{\text{op}} + \|E_1\|_{\text{op}} \leq C\delta(\kappa^{-1} + \gamma^{-1}) = C\delta/\vartheta. \quad (3.3)$$

Here $D_\alpha = \text{diag}(\alpha_b)$ and $D_\xi = \text{diag}(\xi_b)$. This is the cancellation in the proof: the pole error is *not* multiplied by a second channel inverse bound.

For small enough c_0 , $D_\alpha + E_1$ is invertible, with inverse norm at most $2/\alpha_*$. The roots of $\det M(T)$ are the spectrum of

$$Z = (D_\alpha + E_1)^{-1}(D_\alpha D_\xi - E_0), \quad \|Z - D_\xi\|_{\text{op}} \leq C\delta/\vartheta. \quad (3.4)$$

To identify those roots without a hypothesis on the candidate, observe that the leading coefficient is also

$$(L_U U') \text{diag}(\alpha'_b)(V'^\top R_V).$$

Its invertibility forces both outer square factors to be invertible. Thus

$$\det M(T) = \det(L_U U') \det(V'^\top R_V) \prod_b \alpha'_b(T - \xi'_b).$$

In particular Z has exactly the real candidate spectrum, with multiplicity. Lemma 3.1 and (3.4) prove the action bound. Lemma 2.1 proves the pole bound. When $\delta > c_0\vartheta$, compactness of the action interval proves (1.4) after increasing C .

If the channels have full rank, $B = 0$ is equivalent to $\alpha_b(h - \xi_b) = 0$ for every b , by multiplying by the two channel inverses. It is therefore equivalent to all actions being equal. This proves the stated domain of exact identification; the necessity of three clocks follows from Proposition 2.2. Finally,

$$\gamma = \|U \text{diag}(\alpha_b(h - \xi_b))V^\top\|_F \geq \kappa \left(\sum_b \alpha_b^2(h - \xi_b)^2 \right)^{1/2} \geq \frac{1}{2} \alpha_* \Delta \kappa$$

if $\xi_k - \xi_1 \geq \Delta$, because at least one endpoint is at distance $\Delta/2$ from h . The formula for ϑ proves the last assertion. $\square$

Proposition 3.2 (Simple and repeated spectral factors). *For full-rank channels and a nonconstant spectrum, the three matrices determine the spectrum with multiplicity. If the actions are distinct, they also determine U, V, α up to a simultaneous permutation. For a repeated action, its total matrix coefficient is determined; individual rank-one factors within that coefficient need not be.*

Proof. The pole lemma determines h , so the slope A and constant coefficient of $K(T)$ are observed. Take row and column index sets I, J such that A_{IJ} is invertible. Then

$$K(T_1)_{IJ} A_{IJ}^{-1} = U_I \text{diag}(T_1 - \xi_b) U_I^{-1}.$$

For a simple eigenvalue its spectral projector Π_b yields

$$E_b = A_{:J} A_{IJ}^{-1} \Pi_b A_{I:} = \alpha_b U_{:b} V_{:b}^T.$$

Summing all entries gives α_b ; dividing the row and column sums by α_b gives the two stochastic columns. A repeated eigenvalue has a unique projector onto its whole eigenspace, which yields $\sum_{\xi_b=t} \alpha_b U_{:b} V_{:b}^T$ instead. Different positive factorizations of this matrix can give the same curve. The spectrum statement does not require choosing one. $\square$

For example, with three columns in each full-rank channel, the spectra (a, a, b) with $a \neq b$ are covered uniformly as two nearby actions approach a . In contrast, (a, a, a) produces a constant normalized curve; varying a leaves it unchanged. Individual gaps and whole spectrum collapse are mathematically different limits.

4. THE OBSERVATION MODULUS AND HONEST INFERENCE

The next elementary principle isolates what is common to exact fibres and finite-sample uncertainty. Its statistical ingredients are classical; the preceding section supplies the model-specific modulus.

Lemma 4.1 (Residual inversion of an observation quotient). *Let a compact parameter space Θ have continuous observation map F into a finite product of probability simplices and continuous target g in a compact metric space. For a true θ , define*

$$\Omega_\theta(t) = \sup\{d(g(\theta), g(\theta')) : \|F(\theta') - F(\theta)\| \leq t\}.$$

Then $\Omega_\theta(0) = 0$ precisely when the target is constant on the exact observation fibre. In that case $\Omega_\theta(t) \rightarrow 0$ as $t \downarrow 0$. If $e = \|\hat{F} - F(\theta)\|$, a residual minimizer has target error at most $\Omega_\theta(2e)$. The target image of $\{\theta' : \|F(\theta') - \hat{F}\| \leq \eta\}$ has diameter at most $2\Omega_\theta(e + \eta)$, whenever nonempty. On $e \leq \eta$ it contains the true target.

Proof. The first equivalence is the definition at zero. If continuity at zero failed, choose a sequence of competitors at observation distance tending to zero and target distance bounded below; a compact subsequence contradicts the equivalence. A residual minimizer has residual at most e , since the truth is feasible, and the triangle inequality gives distance at most $2e$ from the true observation. Every feasible candidate in the confidence set is at distance at most $e + \eta$ from it. Applying the modulus to each candidate proves the diameter bound. Inclusion of the truth is immediate. $\square$

For (1.1), let $q = m_1 m_2$ and let $\hat{P}_j$ be the empirical matrices. Given $0 < \zeta < 1$, put

$$\eta = \sqrt{\frac{q}{2n} \log \frac{6q}{\zeta}}, \quad \mathcal{C} = \{\omega \in \mathcal{K} : \max_j \|P_\omega(T_j) - \hat{P}_j\|_F \leq \eta\}. \quad (4.1)$$

Report the set $\mathcal{C}_\xi$ of its action spectra. If the set is empty, report model inconsistency and the empty set, not an assigned action. Its diameter is assigned zero only for the loss calculation. Coordinate intervals from the smallest and largest ordered actions in $\mathcal{C}$ give simultaneous bands with the same maximum-coordinate width bound. A minimum-residual estimate is defined over the full compact model.

Theorem 4.2 (Honesty and adaptation without signal floors). *The region (4.1) has coverage at least $1 - \zeta$ uniformly over $\mathcal{K}$, including both singular boundaries. For every true observation P with $s = \vartheta_*(P) > 0$, on the coverage event*

$$\text{diam}_\infty \mathcal{C}_\xi \leq C \min\{1, \eta/s\}.$$

For fixed ζ the expectation and risk estimates (1.6) hold. At $s = 0$ the assertions are coverage and bounded-range losses, not shrinking diameter.

Proof. An entrywise error bound t implies a Frobenius error bound $\sqrt{q}t$. Hoeffding's inequality and a union bound over the $3q$ entries give

$$\Pr\{\max_j \|\hat{P}_j - P_j\|_F > \eta\} \leq 6q \exp(-2n\eta^2/q) = \zeta.$$

For $q = 4$ the conversion is exactly $2t$, not t . Theorem 1.1, applied to a maximizing representative in (1.5), bounds the action modulus by $C \min\{1, t/s\}$. Lemma 4.1 proves coverage and the eventwise diameter bound.

No exceptional event is discarded in the expectation. On every outcome with nonempty $\mathcal{C}$ its diameter is at most $C \min\{1, (e + \eta)/s\}$. Also

$$\mathbb{E}e^2 \leq \sum_{j=1}^3 \mathbb{E}\|\hat{P}_j - P_j\|_F^2 = \frac{1}{n} \sum_{j=1}^3 \left(1 - \sum_{a,b} (P_j)_{ab}^2\right) \leq 3/n.$$

The empty-set convention preserves this bound. Taking expectations proves the diameter assertion. The residual estimator has error bounded by $C \min\{1, e/s\}$ on every outcome; squaring and using the same second moment proves the risk assertion. Compactness gives minimizers and extrema. Measurable choices exist without a selection issue, since the fixed- n categorical sample space is finite. $\square$

Coverage requires independence within each clock. Independence between clocks is used below for the simple fixed-design likelihood argument. Finite forward nets of $\mathcal{K}$ give an exhaustive implementation: if the mesh in parameter space is r , uniform forward Lipschitz continuity allows padding the residual radius by Cr and the action bands by r . The preceding proofs then replace η by $\eta + Cr$. Neither inverse conditioning nor a guessed latent rank floor is needed; the capacity k is part of the declared experiment.

5. PRODUCT SINGULARITIES IN EVERY LATENT RANK

The following family exposes the loss which remains after both observed marginals have been held fixed. It also shows why a product rather than an individual channel weakness occurs.

Let $u_0 = m_1^{-1}\mathbf{1}$ and $v_0 = m_2^{-1}\mathbf{1}$. Choose unit zero-sum vectors $e \in \mathbb{R}^{m_1}$, $f \in \mathbb{R}^{m_2}$. If $k > 2$, choose orthonormal zero-sum vectors $e_3, \dots, e_k$ orthogonal to e , and $f_3, \dots, f_k$ orthogonal to f . Their existence follows from $m_i \geq k$. Fix a small $c > 0$. Choose distinct interior actions, with the first two $\xi_1 = s_0 - d$, $\xi_2 = s_0 + d$, and all others strictly above ξ_2 and separated from this

pair. Take $\alpha_b = 1/k$. For $0 \leq t \leq t_0 < d/2$, put $\lambda = 1 - t/d$ and

$$\begin{aligned} \xi_{1,t} &= \xi_1 + t, & \xi_{2,t} &= \xi_2 - t, & \xi_{b,t} &= \xi_b \quad (b \geq 3), \\ U_{t,1} &= u_0 + \frac{\epsilon_u e}{2\lambda}, & U_{t,2} &= u_0 - \frac{\epsilon_u e}{2\lambda}, & U_{t,b} &= u_0 + c e_b \quad (b \geq 3), \\ V_{t,1} &= v_0 + \frac{\epsilon_v f}{2\lambda}, & V_{t,2} &= v_0 - \frac{\epsilon_v f}{2\lambda}, & V_{t,b} &= v_0 + c f_b \quad (b \geq 3). \end{aligned} \quad (5.1)$$

Take $0 \leq \epsilon_u, \epsilon_v \leq \epsilon_*$ with ϵ_* sufficiently small. All channel and joint cell probabilities have positive lower bounds independent of $t, \epsilon_u, \epsilon_v$. The pole h is constant along the family.

Lemma 5.1 (Exact marginal cancellation). *Both readout marginals in (5.1) are independent of t at every clock in $[T_1, T_3]$, and*

$$P_t(T) - P_0(T) = \frac{T - s_0}{2k(T - h)} \epsilon_u \epsilon_v (\lambda^{-2} - 1) e f^\top. \quad (5.2)$$

For positive contrasts,

$$\kappa(\omega_t) = \frac{\epsilon_u \epsilon_v}{2\lambda^2}, \quad \vartheta_*(\mathcal{P}(\omega_t)) = \vartheta(\omega_t) \asymp \epsilon_u \epsilon_v. \quad (5.3)$$

The observation distance is comparable to $\epsilon_u \epsilon_v t$. If either contrast is zero, all these laws are exactly equal although the spectrum changes.

Proof. In the pair contribution, expand

$$(T - s_0 + \lambda d)(u_0 + \epsilon_u e/(2\lambda))(v_0 + \epsilon_v f/(2\lambda))^\top + (T - s_0 - \lambda d)(u_0 - \epsilon_u e/(2\lambda))(v_0 - \epsilon_v f/(2\lambda))^\top.$$

Its constant part is $2(T - s_0)u_0 v_0^\top$, its linear parts are $d\epsilon_u e v_0^\top + d\epsilon_v u_0 f^\top$, and its bilinear part is $(T - s_0)\epsilon_u \epsilon_v e f^\top/(2\lambda^2)$. Divide by $k(T - h)$ and subtract $t = 0$ to obtain (5.2). Zero sums of e, f prove the marginal assertion directly.

The vector $(1, -1, 0, \dots, 0)/\sqrt{2}$ is a right singular vector of U_t with singular value $\epsilon_u/(\sqrt{2}\lambda)$. On its orthogonal complement the map is independent of ϵ_u and has a positive smallest singular value: the independent output directions are $u_0, e_3, \dots, e_k$. These directions are orthogonal to e . Choosing ϵ_* below that fixed singular value gives the stated exact smallest singular value. The same argument applies to V_t , proving the first identity in (5.3). Taking marginals of B shows that its e and f components have magnitudes proportional to ϵ_u and ϵ_v , respectively; hence $\gamma \geq c \max(\epsilon_u, \epsilon_v)$. Consequently $\vartheta \asymp \epsilon_u \epsilon_v$. All spectra in this family are simple and the channels are full rank when both contrasts are positive. Proposition 3.2 therefore gives $\vartheta_* = \vartheta$. Finally $T - s_0$ and $T - h$ have uniform positive bounds, and $\lambda^{-2} - 1 \asymp t$ on the fixed interval. This proves the distance comparison and the exact singular statement. $\square$

The vanishing interaction is the tensor product of the two contrast directions. There are no first-order marginal changes left to exploit. The calculation is not restricted to binary alphabets, square matrices, or two latent components.

6. COLLAPSE OF THE WHOLE PROJECTIVE SPECTRUM

Fix strictly positive full-column-rank channels, positive weights, and distinct real numbers z_b with $\sum_b \alpha_b z_b = 0$. Take an interior a and sufficiently small $s \geq 0$. Consider

$$\xi_b(t) = a + t + s z_b, \quad 0 \leq t \leq t_0, \quad (6.1)$$

with t_0 fixed so all actions remain in (L, D) . Here $h_t = a + t$ and

$$A = U \operatorname{diag}(\alpha_b) V^\top, \quad B = -s U \operatorname{diag}(\alpha_b z_b) V^\top$$

are independent of t . Therefore

$$P_t(T) - P_0(T) = \frac{tB}{(T-a-t)(T-a)}. \quad (6.2)$$

For $s > 0$, $\gamma \asymp s$, κ is bounded below, and $\vartheta_* = \vartheta \asymp s$, by simple-spectrum uniqueness. The observation distance is comparable to st , whereas the action distance is exactly t . At $s = 0$ the laws are constant in both T and t , although the common action changes. Thus the contrast term in Theorem 1.1 is necessary even for perfectly well-conditioned channels. This is a second singular boundary, not the channel-rank boundary in Section 5.

7. SHARP STATISTICAL ORDERS AND ADAPTIVE CLOCKS

Theorem 7.1 (Minimax transition and honest-diameter lower bounds). *For each fixed admissible k, m_1, m_2 there is $s_* > 0$ such that, for $0 < s \leq s_*$ and $N = 3n$,*

$$c \min\{1, (Ns^2)^{-1}\} \leq \inf_{\hat{\xi}} \sup_{\omega: \vartheta_*(P(\omega)) \geq s} \mathbb{E}_\omega d_\infty(\hat{\xi}, \xi)^2 \leq C \min\{1, (Ns^2)^{-1}\}. \quad (7.1)$$

The upper bound uses the balanced three-clock design. The lower bound remains valid if the infimum also includes arbitrary randomized, predictable choices of each of N clocks in $[T_1, T_3]$.

Let a random spectrum region J be honest at level $1 - \zeta$ over $\mathcal{K}$, with $0 < \zeta < 1/4$. At the base point of either positive family in Sections 5 and 6,

$$\mathbb{E}_0 \text{diam}_\infty J \geq c_\zeta \min\{1, (\sqrt{N}\vartheta_*(P_0))^{-1}\}. \quad (7.2)$$

This holds for any such clock strategy. At the exact singular limits of these families there are nonvanishing risk and honest-diameter lower bounds. Equal channel contrasts ϵ give the transition $N\epsilon^4 \asymp 1$; full-spectrum spread s with fixed channels gives $Ns^2 \asymp 1$.

Proof. The upper bound follows from Theorem 4.2. In either lower-bound family write $\beta = \epsilon_u \epsilon_v$ or $\beta = s$, respectively. The exact identities and uniform positive cell floors give, uniformly for $T \in [T_1, T_3]$,

$$\text{KL}(P_t(T), P_0(T)) \leq \sum_{a,b} \frac{((P_t)_{ab} - (P_0)_{ab})^2}{(P_0)_{ab}} \leq C\beta^2 t^2.$$

For a predictable randomized clock strategy, condition on the entire past and the internal randomization at each step. The policy has the same conditional rule under both parameters, and only the next categorical law contributes to relative entropy. The chain rule therefore bounds the entropy of the complete transcript by $CN\beta^2 t^2$. This includes fixed and uneven clock allocations.

Choose $t = c_1 \min\{t_0, (\sqrt{N}\beta)^{-1}\}$ with c_1 small. Pinsker's inequality makes the total variation distance of the transcript laws at most $1/2$. The spectra are distance t apart. Choosing the nearer spectrum converts any estimator into a test, so its larger squared risk is at least $t^2/16$.

To obtain the specified class in (7.1), use the product family with $\epsilon_u = \epsilon_v$ and choose their product to be Ms for a fixed sufficiently large M . Uniformly on the small t interval, Lemma 5.1 gives $\vartheta_* \geq c\epsilon_u \epsilon_v \geq s$. The upper contrast restriction holds for $s \leq s_*$. The two-point bound is then the lower bound in (7.1). The collapse family supplies the second attained singular order as well.

For an honest set J , transfer coverage under the alternative to the base law. Then

$$\Pr_0\{\xi(0), \xi(t) \in J\} \geq 1 - 2\zeta - \text{TV}(\mathbb{P}_t, \mathbb{P}_0).$$

Decrease c_1 , depending on ζ , so the last expression is bounded below by a positive constant. On this event the diameter is at least t , which proves (7.2). At a zero contrast, or at zero full-spectrum spread, choose fixed distinct t ; the exact identities make the transcript laws identical for every policy. The same testing and coverage arguments give the nonvanishing limits. $\square$

The theorem is a uniform order statement and gives pointwise lower bounds on the displayed families. It does not identify the exact directional condition number of every anisotropic latent factorization. In particular a small product computed from an arbitrary repeated-spectrum representative is not declared to be an intrinsic obstruction; (1.5) removes precisely that dependence on representation.

8. RELATION TO THE OTHER OBSERVATION QUOTIENTS

The result reorganizes two earlier mechanisms: the arbitrary-rank projective clock pencil and the binary determinant-product modulus. For $k = m_1 = m_2 = 2$, stochastic columns imply

$$\frac{|\det U|}{\sqrt{2}} \leq \sigma_2(U) \leq |\det U|, \quad \frac{|\det V|}{\sqrt{2}} \leq \sigma_2(V) \leq |\det V|.$$

Indeed $1 \leq \sigma_1(U), \sigma_1(V) \leq \sqrt{2}$, and the product of singular values is the absolute determinant. On the separated action/reference class, Theorem 1.1 thus gives the same single inverse power of $|\det U \det V|$. Also $\alpha_1 = (\xi_2 - h)/(\xi_2 - \xi_1)$, so a fixed action/reference gap recovers the earlier throughput estimate. No determinant of a higher-degree scalar polynomial or gap-dependent root derivative is needed for the general spectrum theorem.

The shared two-clock spatial experiment has a different nuisance space: its clock factors are common across sites. Spatial calibration can then replace a third clock. The exact shared theorem, including its one-clock geometric alternatives, is retained in the companion. Its local correctly-labelled calibration estimate is not a global noise bound through profile collisions. The new theorem supplies a singular theory for the projective affine experiment; it does not silently transfer its modulus to that different shared experiment.

Likewise, the polynomial, rational and compact-surface detector quotients retain their own secant and tangent tests and clock designs. They are preserved in full, together with the branched-clock version of the unchanged logarithmic law. Their divisor counts remain valid sufficient counts. The sharp count here is governed by the affine numerator and one projective normalization parameter, not by a claim that an ambient meromorphic pole budget is always optimal.

The companion also retains finite-record geometric certificates, component coverings, return-lift constructions, and raw-acquisition laws. Metric rigidity there is supplied by the stated classical boundary-rigidity theorems. The present theorem is an upstream observation result: it adds neither a partial-data boundary-rigidity claim nor a new geometric hypothesis. Separating these logical roles allows the action inverse and its singular statistical transition to form a self-contained theorem, while retaining the other mathematical regimes with all their assumptions.

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